

C2
18.10.2014

THE UNIFIED PROCESS

4 phases in the UP

- inception
- elaboration
- construction
- transition

↑ requirements & important in architecture project

each phase has a # of iterations. Each iteration produces a new increment.

Agile Methods

- evolution of the incremental model

Principles

- customer involvement
- incremental delivery
- people not process
- embrace change
- maintain simplicity

Extreme Programming

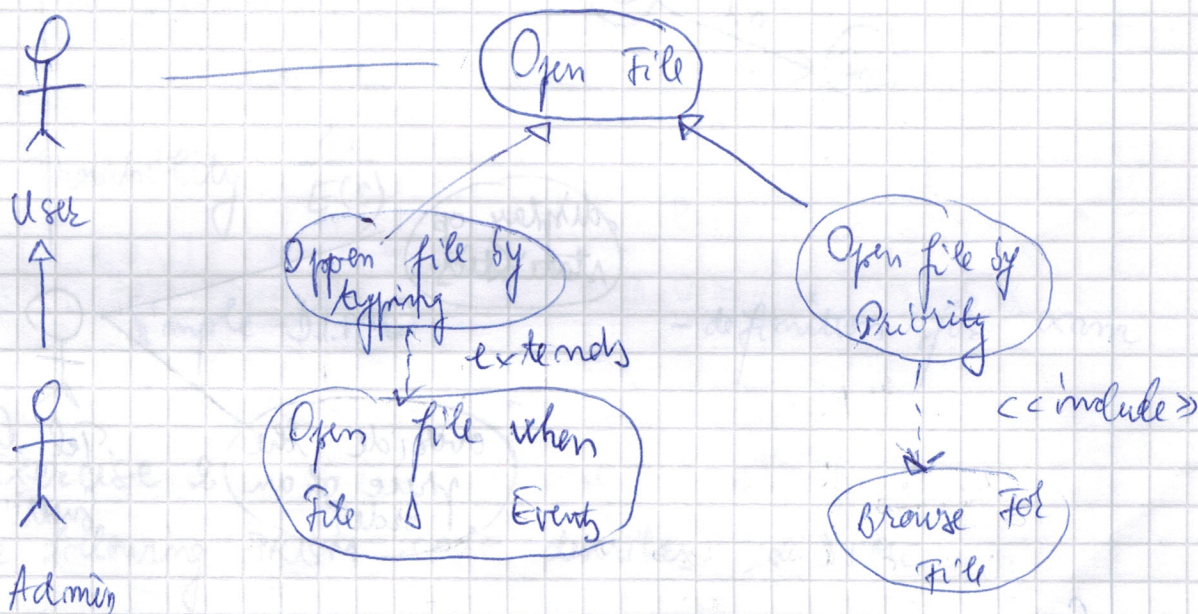
- all requirements are expressed as user stories (scenarios)

BASIC SOFTWARE DEVELOPMENT ACTIVITIES

Analysis and specification design

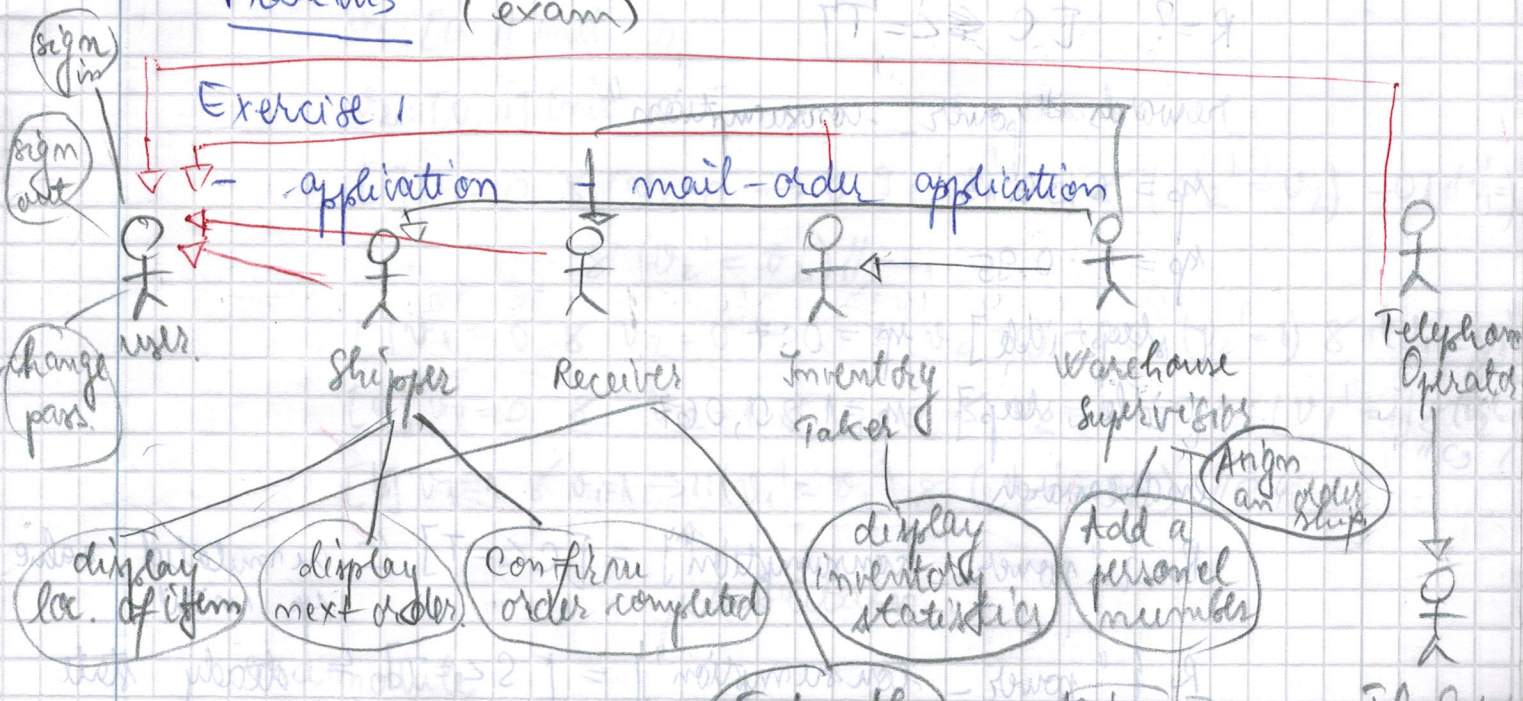
Cs

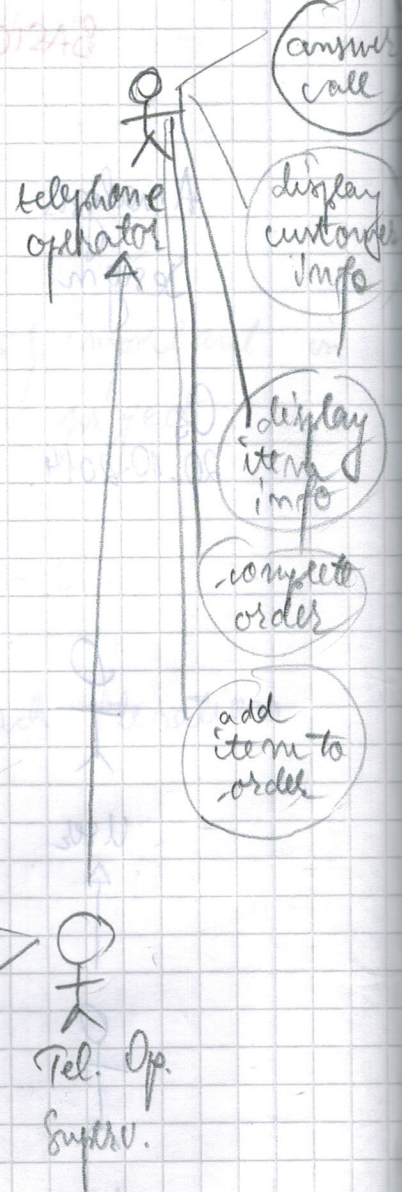
20.10.2014.



Problems (exam)

Exercise 1





C9.

$R = ?$ [rewardprop]

$R = ?$ [$C \leq T$]

rewards "power consumption"

$\mu_p = 0 : 0.13$

$\mu_p = 1 : 0.95$

[sleep & idle] $\mu_p = 0 : 7$

[idle & sleep] $\mu_p = 1 : 0.067$

end rewards

$R \uparrow$ "power consumption" = [$C \leq T$] $\Leftarrow$ accumulated value up until

$R \uparrow$ "power consumption" = [$S \leq T$] $\Leftarrow$ steady state

C5.

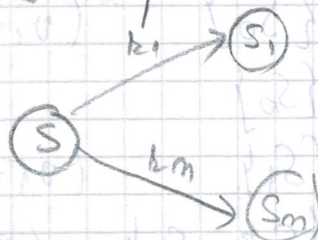
3.11.2014

Discrete - Time Markov Chains

Continuous - Time Markov Chains

Two interpretations of a CTMC.

$$E(s) = \lambda_1 + \dots + \lambda_m$$

Probability $\frac{\lambda_i}{E(s)}$

Simple JTMC

- definition for exam

Exercise 2.

The following PRISM code describes a JTMC

dtmc

module M

 $v_1: [0..1]$ init 0, $v_2: [0..1]$ init 0,
 $[] v_1 = 0 \ \& \ v_2 = 0 \rightarrow 0.9 : (v_1' = 0) \ \& \ (v_2' = v_2) + 0.1 : (v_1' = 1) \ \& \ (v_2' = v_2) \parallel c_1$
 $[] v_1 = 0 \ \& \ v_2 = 0 \rightarrow 0.7 : (v_2' = 0) + 0.3 : (v_2' = 1) \ \& \ (v_1' = v_1) \parallel c_2$
 $[a] v_1 = 0 \ \& \ v_2 = 1 \ \& \ v_1 = 0 \ \& \ v_2 = 1 \rightarrow 1 : (v_1' = 1) \ \& \ (v_2' = 1) \parallel c_3$
 $[b] v_1 = 1 \ \& \ v_2 = 1 \rightarrow 1 : (v_1' = v_1) \ \& \ (v_2' = v_2) \parallel c_4$

↑

probabilities

endmodule

Possible system states

$$S_0 = (0, 0)$$

$$S_1 = (0, 1)$$

$$S_2 = (1, 0)$$

$$S_3 = (1, 1)$$

$$S_{C1} = \{S_0\} \quad (v_1=0, v_2=0 \text{ deci } S_0)$$

$$S_{C2} = \{S_0\}$$

$$S_{C3} = \{S_1\} \quad (v_1=0, v_2=1 \text{ deci } S_1)$$

$$S_{C4} = \{S_2, S_3\} \quad (v_2 \text{ poate fi } < 1)$$

$$\mu_{C1, S_0}(S_0) = 0.9$$

$$\mu_{C1, S_0}(S_2) = 0.1$$

$$\Rightarrow \mu_{C1, S_0} = [S_0 \rightarrow 0.9, S_2 \rightarrow 0.1]$$

$$\mu_{C2, S_0}(S_0) = 0.7$$

$$\mu_{C2, S_0}(S_1) = 0.3$$

$$\Rightarrow \mu_{C2, S_0} = [S_0 \rightarrow 0.7, S_1 \rightarrow 0.3]$$

$$\mu_{C3, S_1}(S_3) = 1 \Rightarrow \mu_{C3, S_1} = [S_3 \rightarrow 1]$$

$$\mu_{C4, S_2}(S_2) = 1 \Rightarrow \mu_{C4, S_2} = [S_2 \rightarrow 1]$$

$$\mu_{C4, S_3}(S_3) = 1 \Rightarrow \mu_{C4, S_3} = [S_3 \rightarrow 1]$$

$$\bar{\varphi} =$$

	S_0	S_1	S_2	S_3
S_0	0.9+0.1=1	0.3	0.1	0
S_1	0	0	0	1
S_2	0	0	1	0
S_3	0	0	0	1

from S_0 to S_2

$P =$

	S_0	S_1	S_2	S_3
S_0	0.8	0.15	0.05	0
S_1	0	0	0	1
S_2	0	0	1	0
S_3	0	0	0	1

$$\frac{1.6}{(1.6 + 0.3 + 0.1)} = 0.8$$

$$\frac{0.3}{2} = 0.15$$

se normalizăm pe rânduri

C5

10.11.2014

ctmc

module M

$V_1: [0..1]$ unit 0;

$V_2: [0..1]$ unit 0;

[] $V_1=0 \ \& \ V_2=0 \rightarrow 4.5 : (V_1'=0) \ \& \ (V_2'=V_2) +$
 $0.5 : (V_2'=1) \ \& \ (V_2'=V_2) // = c_1$

[] $V_1=0 \ \& \ V_2=0 \rightarrow 3.5 : (V_2'=0) \ \& \ (V_1'=V_1) + 1.5 :$
 $(V_2'=1) \ \& \ (V_1'=V_1) // = c_2$

[a] $V_1=0 \ \& \ V_2=1, \ \& \ V_1=0 \ \& \ V_2=1 \rightarrow (V_1'=1) \ \&$
 $\& \ \text{true} // = c_3$
 $//$
 $(V_2'=V_2)$

[5] $V_1=1 \ \& \ V_1=1 \rightarrow 2 : \text{true} \ \& \ \text{true} // c_4$
 $//$
 $(V_1'=V_1) \ \& \ (V_2'=V_2)$

end module.

$$\mu_{c_1, S_0}(S_0) = 4.5$$

$$\mu_{c_1, S_0}(S_2) = 0.5$$

$$\mu_{c_2, S_0}(S_0) = 3.5$$

$$\mu_{c_3, s_1}(s_3) = 1$$

$$\mu_{c_4, s_3}(s_3) = 2$$

$$\mu_{c_4, s_2}(s_2) = 2$$

! exam: 3 types of automaton presented
(reproduce the definitions)
what is a dtmc, ctmc, mdp.

$R =$

	s_0	s_1	s_2	s_3
s_0	8	1.5	0.5	0
s_1	0	0	0	1
s_2	0	0	2	0
s_3	0	0	0	2

ctmc.

module M

$V_1 = [0..1]$ init 0;

$V_2 = [0..1]$ init 0;

[] $V_1 = 0 \ \& \ V_2 = 0 \rightarrow 0.9 : (V_1' = 0) \ \& \ (V_2' = V_2) +$
 $0.1 : (V_1' = 1) \ \& \ (V_2' = V_2) \quad \parallel = c_1$

[] $V_1 = 0 \ \& \ V_2 = 0 \rightarrow 0.7 : (V_2' = 0) \ \& \ (V_1' = V_1) +$
 $0.3 : (V_2' = 1) \ \& \ (V_1' = V_1) \quad \parallel = c_2$

[a] $V_1 = 0 \ \& \ V_2 = 1 \ \& \ V_1 = 0 \ \& \ V_2 = 1 \rightarrow 1 : (V_1' = 1)$
 $\ \& \ \text{true} \quad \parallel = c_3$

[b] $V_1 = 1 \ \& \ V_1 = 1 = \text{true} \ \& \ \text{true} \quad \parallel = c_4$

$$\left. \begin{array}{l} \mu_{c_1, s_0}(s_0) = 0.9 \\ \mu_{c_1, s_0}(s_1) = 0.1 \end{array} \right\} \Rightarrow \mu_{c_1, s_0} = [s_0 \rightarrow 0.9, s_2 \rightarrow 0.1]$$

$$\left. \begin{array}{l} \mu_{c_2, s_0}(s_0) = 0.7 \\ \mu_{c_2, s_0}(s_1) = 0.3 \end{array} \right\} \Rightarrow \mu_{c_2, s_0} = [s_0 \rightarrow 0.7, s_1 \rightarrow 0.3]$$

$$\mu_{c_3, s_1}(s_3) = 1 \Rightarrow \mu_{c_3, s_1} = [s_3 \rightarrow 1]$$

$$\mu_{c_4, s_3}(s_3) = 1 \Rightarrow \mu_{c_4, s_3} = [s_3 \rightarrow 1]$$

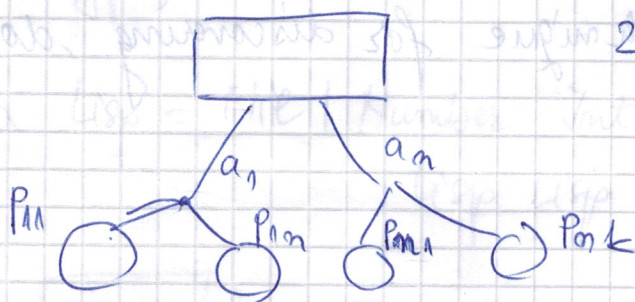
$$\mu_{c_4, s_2}(s_2) = 1 \Rightarrow \mu_{c_4, s_2} = [s_2 \rightarrow 1]$$

$$\text{Steps}(s_0) = \{ \mu_{c_1, s_0}, \mu_{c_2, s_0} \} = \{ [s_0 \rightarrow 0.9, s_2 \rightarrow 0.1], [s_0 \rightarrow 0.7, s_1 \rightarrow 0.3] \}$$

$$\text{Steps}(s_1) = \{ \mu_{c_3, s_1} \} = \{ [s_3 \rightarrow 1] \}$$

$$\text{Steps}(s_2) = \{ \mu_{c_4, s_2} \} = \{ [s_2 \rightarrow 1] \}$$

$$\text{Steps}(s_3) = \{ \mu_{c_4, s_3} \} = \{ [s_3 \rightarrow 1] \}$$



$$2^X = P(X) = \{ Y \mid Y \subseteq X \}$$

$$p_{m1} + \dots + p_{mk} = 1$$

$$p_{m1} + \dots + p_{mn} = 1$$

$$c) \text{ Steps } (s_0) = \{(\uparrow, \mu_{c1}, s_0), (\uparrow, \mu_{c2}, s_0)\} \\ = \{(\uparrow, [s_0 \rightarrow 0.9, s_2 \rightarrow 0.1]), (\uparrow, [s_0 \rightarrow 0.7, s_1 \rightarrow 0.3])\}$$

$$\text{Steps } (s_1) = \{(a, \mu_{c3}, s_1)\} = \{(a, [s_3 \rightarrow 1])\}$$

$$\text{Steps } (s_2) = \{(b, \mu_{c4}, s_2)\} = \{(b, [s_2 \rightarrow 1])\}$$

$$\text{Steps } (s_3) = \{(b, \mu_{c4}, s_3)\} = \{(b, [s_3 \rightarrow 1])\}$$

! Nov 27 - Homework
- experimente reale de la proiect.

5.9 The Process of Developing Class Diagrams

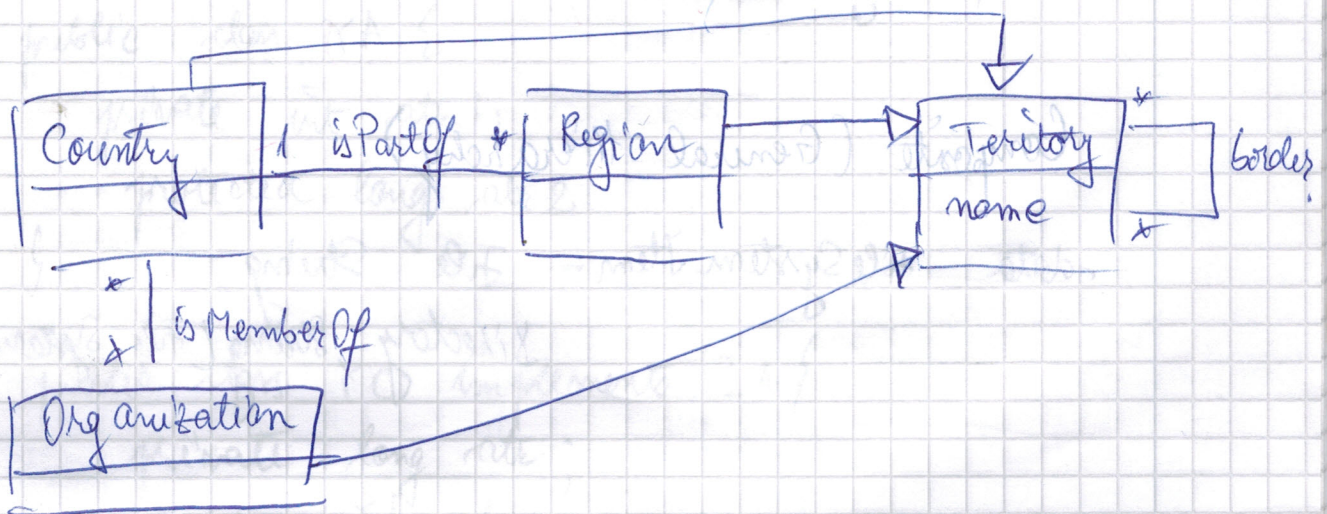
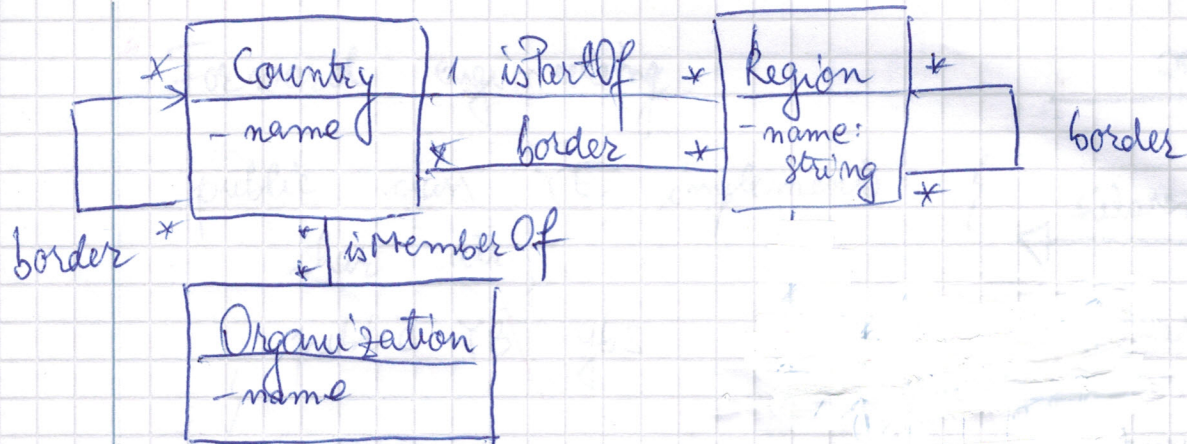
Week 8
24.11.2014

closing 5.

A simple technique for discovering domain classes.

Exercise 3

(E91 - L05)

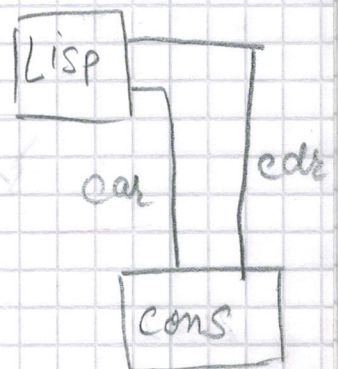
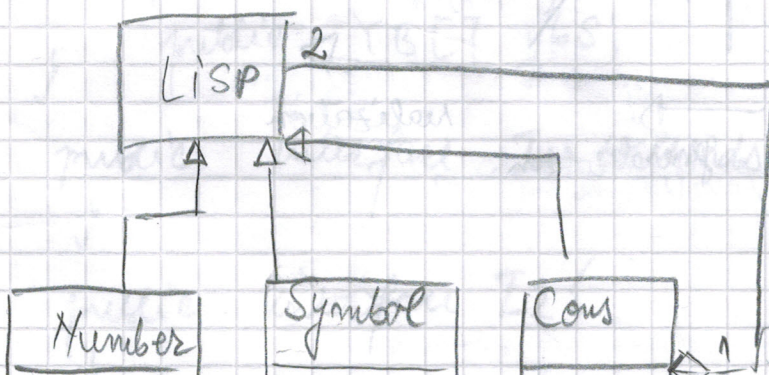


Add the superclass territory.

Exercise:

Use the composite design pattern to model the Lisp data type.

data Lisp = Nil | Number Int | Symbol String | Cons
Lisp Lisp



anac

generaliz

realiz

dependency

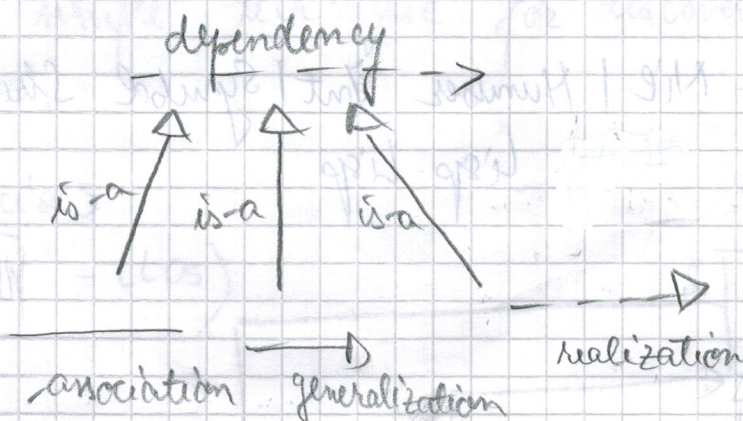
Composite (General Hierarchy)

data FileSystemItem - File String

Directory String [File System Item]

Partial - & decumbie

Forward Engineering
UML → JAVA



Forward engineering

```
public class YE implements {  
    char atr;
```

```
    public YB ybs  
}
```

```
public class YA {  
    private int atr1;  
    protected long atr2;  
}
```

```
public class YC implements IA {  
    private long atr;  
    public YD yd;  
}
```

```
public class YB {  
    public YD[] yds,  
    public void op1 () {}  
    public void op2 () {}  
}
```

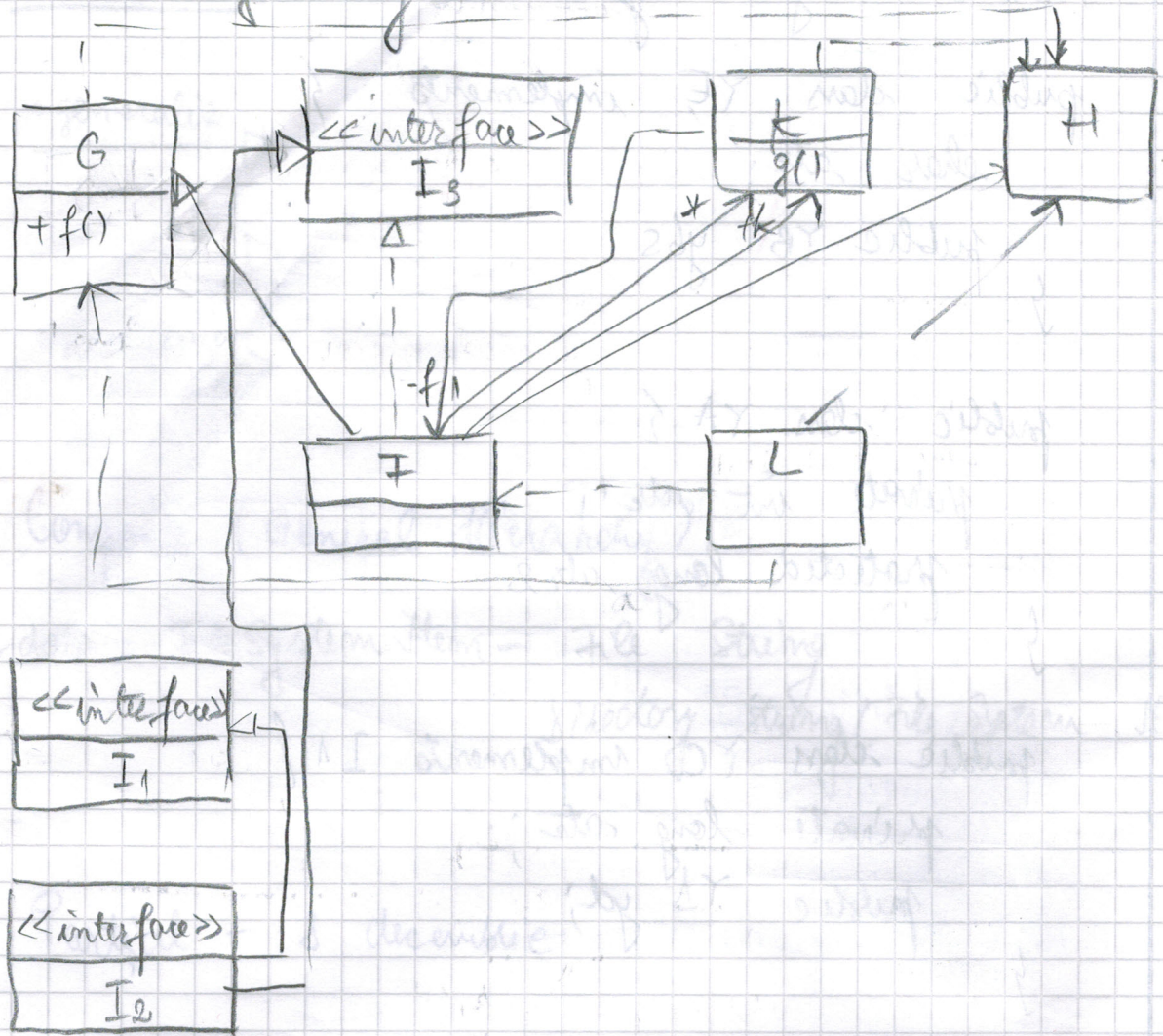
```
public class YD {  
    public int atr;
```

```
    public YB[] ybs;  
}
```

```
public interface I1 extends I2, I3 {
```

```
    public interface I2 {
```


Reverse engineering



Recuperare

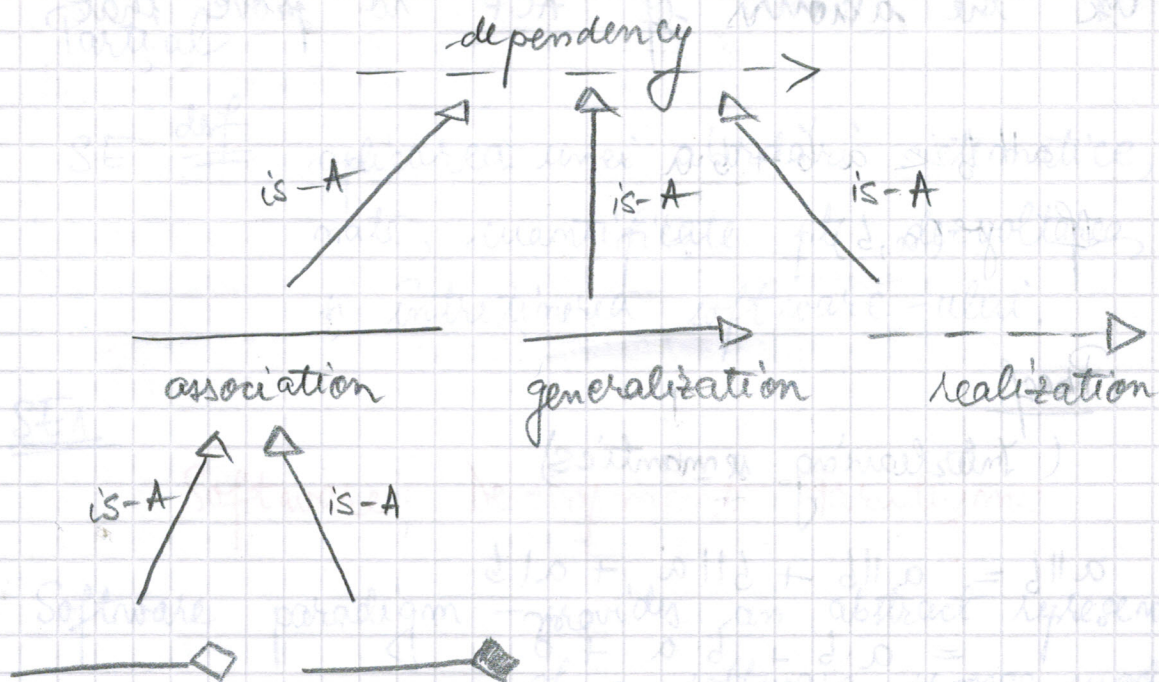
26.11.2014

prism - 5 slides

DTMC, CTMC, MSP

Prism - specific language

ex: definite DTMC

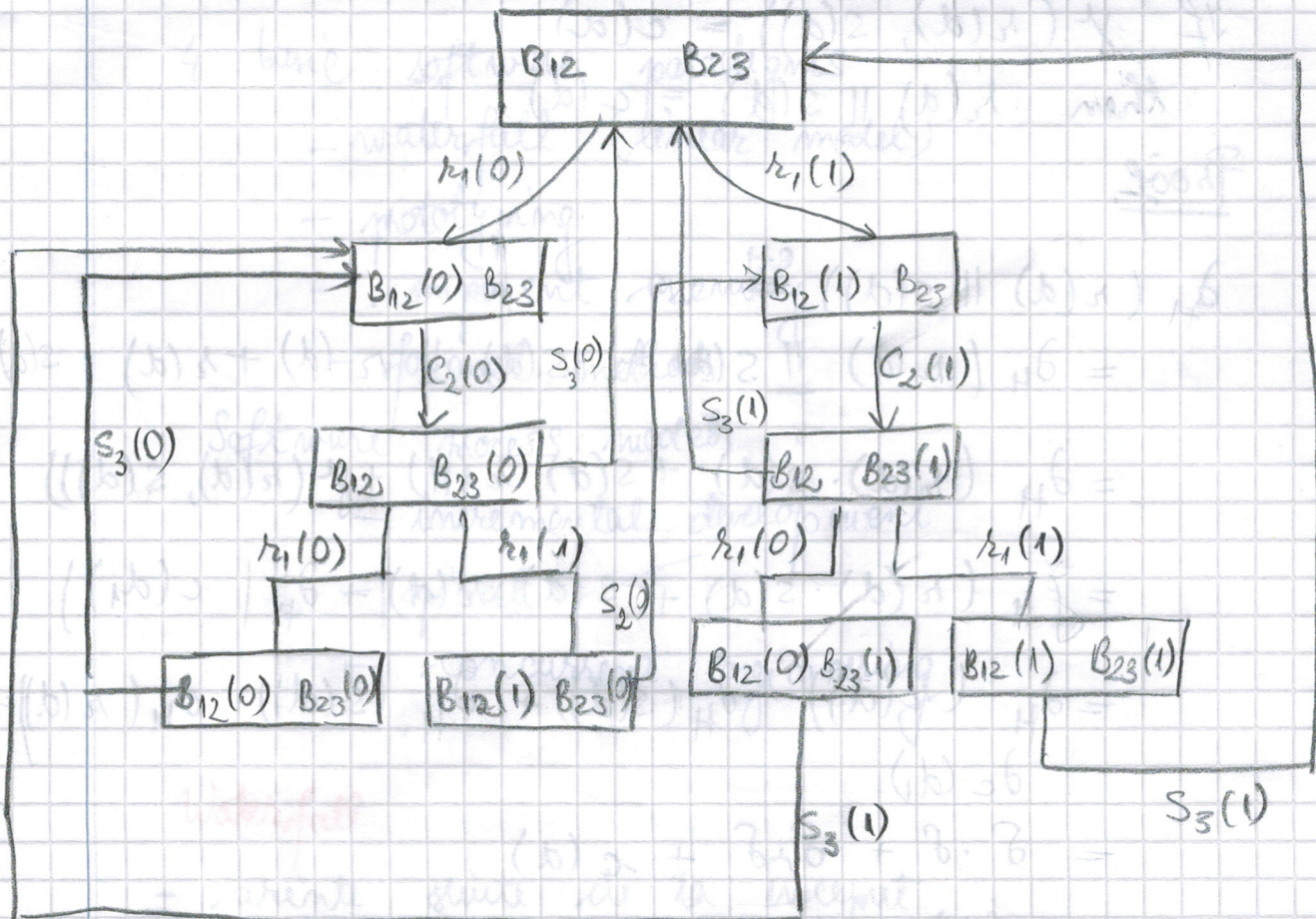


Exercise 1

$c_j(b) \cdot r_j(b)$

$s_j(b)$

HAXIS STAKE



$$I = \{ h_2(b), s_2(b) \mid b = 0, 1 \}$$

Use the axioms of ACP to prove that

$$a \parallel b \leq a \cdot b + b \cdot a$$

$$\text{If } \gamma(a, b)$$

Proof

(Interleaving semantics)

$$\begin{aligned} a \parallel b &= a \parallel b + b \parallel a + a \cdot b \\ &= a \cdot b + b \cdot a + \delta \\ &= a \cdot b + b \cdot a \end{aligned}$$

$$\begin{aligned} \text{If } \gamma(r(d), s(d)) &= c(d) \\ \text{then } r(d) \parallel s(d) &= c(d) \end{aligned}$$

Proof

$$\begin{aligned} \partial_H(r(d) \parallel s(d)) &\stackrel{\text{CH}}{=} \\ &= \partial_H(r(d) \parallel s(d) + s(d) \parallel r(d) + r(d) \cdot s(d)) \\ &= \partial_H(r(d) \cdot s(d) + s(d) \cdot r(d) + \gamma(r(d), s(d))) \\ &= \partial_H(r(d) \cdot s(d) + s(d) \cdot r(d) + \partial_H \mid c(d)) \\ &= \partial_H(r(d)) \cdot \partial_H(s(d)) + \partial_H(s(d)) \cdot \partial_H(r(d)) + \\ &\quad \partial_{c(d)} \\ &= \delta \cdot \delta + \delta \cdot \delta + c(d) \\ &= c(d) \end{aligned}$$